AIM:To study and implement Programming menus, dialog, dialog fragments

1. Options Menu
2. Context Menu
3. Alert Dialog
4. Dialog Fragment

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"

    android:gravity="center"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <Button

        android:id="@+id/btnDialog"

        android:text="Open Alert Dialog"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"/>

    <Button

        android:id="@+id/btnFragmentDialog"

        android:text="Open Dialog Fragment"

        android:layout_marginTop="20dp"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"/>

    <TextView
```

```
    android:id="@+id/txtContext"
    android:text="Long Press Me (Context Menu)"
    android:textSize="18sp"
    android:padding="20dp"
    android:layout_marginTop="30dp"
    android:background="#DDDDDD"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
</LinearLayout>
```

Create Menu Resource

res → menu → create **menu_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item
      android:id="@+id/menu_about"
      android:title="About"/>
   <item
      android:id="@+id/menu_exit"
      android:title="Exit"/>
</menu>
```

MainActivity.java

```
package com.example.menudialogdemo;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.DialogFragment;
import android.os.Bundle;
import android.view.ContextMenu;
```

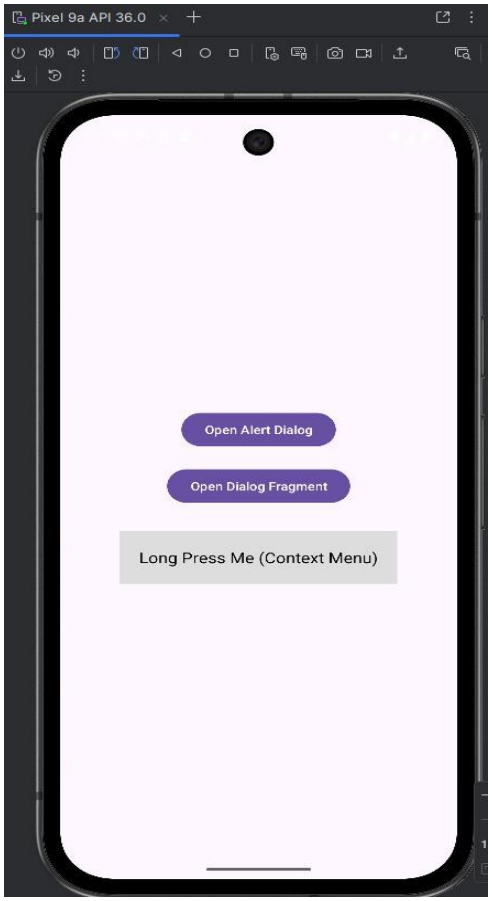
```
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
Button btnDialog, btnFragmentDialog;
   TextView txtContext;
@Override
  protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
btnDialog = findViewById(R.id.btnDialog);
        btnFragmentDialog = findViewById(R.id.btnFragmentDialog);
        txtContext = findViewById(R.id.txtContext);
// Register Context Menu
        registerForContextMenu(txtContext);
        // Alert Dialog
        btnDialog.setOnClickListener(v -> {
            AlertDialog.Builder builder = new AlertDialog.Builder(this);
            builder.setTitle("Alert")
                    .setMessage("This is Alert Dialog")
                    .setPositiveButton("OK", (dialog, which) ->
                            Toast.makeText(this, "OK Clicked", Toast.LENGTH_SHORT).show())
                    .setNegativeButton("Cancel", null)
                    .show();
 });
        // Dialog Fragment
```

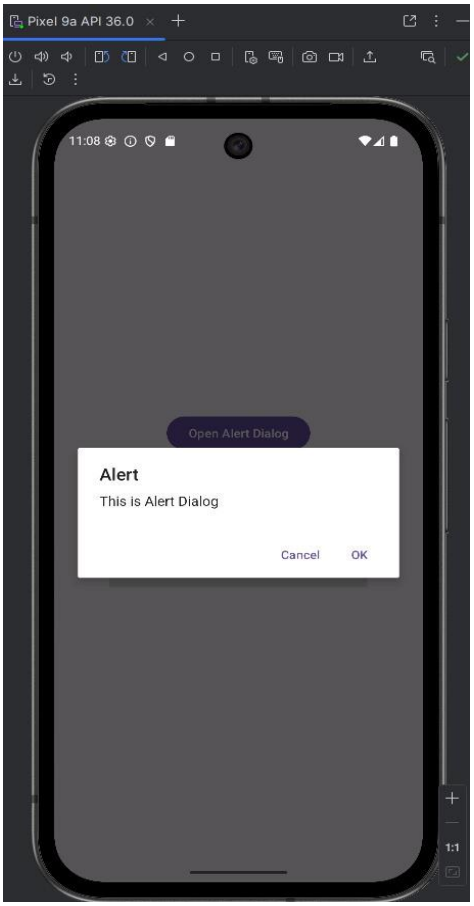
```
        btnFragmentDialog.setOnClickListener(v -> {
            DialogFragment dialog = new MyDialogFragment();
            dialog.show(getSupportFragmentManager(), "MyDialog");
        });
    }
    // Options Menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        if (item.getItemId() == R.id.menu_about) {
            Toast.makeText(this, "Menu: About", Toast.LENGTH_SHORT).show();
            return true;
        } else if (item.getItemId() == R.id.menu_exit) {
            finish();
            return true;
        }
        return super.onOptionsItemSelected(item);
    }
    // Context Menu
    @Override
    public void onCreateContextMenu(ContextMenu menu, View v,
                                    ContextMenu.ContextMenuInfo menuInfo) {
        super.onCreateContextMenu(menu, v, menuInfo);
        menu.setHeaderTitle("Context Menu");
        menu.add(0, 1, 0, "Edit");
```

```
        menu.add(0, 2, 0, "Delete");
    }
    @Override
    public boolean onContextItemSelected(MenuItem item) {
        if (item.getItemId() == 1) {
            Toast.makeText(this, "Edit Selected", Toast.LENGTH_SHORT).show();
        } else if (item.getItemId() == 2) {
            Toast.makeText(this, "Delete Selected", Toast.LENGTH_SHORT).show();
        }
        return true;
    }
}
```

Create Dialog Fragment Class

```
package com.example.menudialogdemo;
import android.app.Dialog;
import android.os.Bundle;
import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.DialogFragment;
public class MyDialogFragment extends DialogFragment {
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        return new AlertDialog.Builder(requireActivity())
                .setTitle("Dialog Fragment")
                .setMessage("This is Dialog Fragment Example")
                .setPositiveButton("OK", null)
                .create();
    }
```


Pixel 9a API 36.0
Open Alert Dialog
Open Dialog Fragment
Long Press Me (Context Menu)


Pixel 9a API 36.0
11:08
Open Alert Dialog
Alert
This is Alert Dialog
Cancel
OK


Pixel 9a API 36.0
11:08
Open Alert Dialog
Dialog Fragment
This is Dialog Fragment Example
OK